RESEARCH ARTICLE

Feedback-aware calibration of traffic prediction intervals under delayed and missing sensor reports

Shishir Singh Chauhan[1] and Satyabrata Roy[1*]

[1]Department of Computer Science and Engineering, Manipal University Jaipur, Jaipur, 303007, Rajasthan, India.

*Corresponding author(s). E-mail(s): satyabrata.roy@jaipur.manipal.edu;
Contributing authors: shishir.chauhan@jaipur.manipal.edu;

Abstract

Traffic prediction intervals rely on forecast errors whose reports may arrive late or disappear during sensor outages. We investigate how to calibrate uncertainty when both predictor inputs and outcome feedback are impaired. The proposed signed-residual method combines spatial pooling, context matching, target-age weighting and historical fallback. A locked causal replay separates input availability from feedback release and compares calibrators on frozen forecasting backbones. Experiments use traffic speeds from Los Angeles (METR-LA), the San Francisco Bay Area (PEMS-BAY), and Simulation of Urban MObility (SUMO) episodes. The registered joint interval-score and coverage rule was met in 12 of 16 real-data backbone–condition cells. With the Graph WaveNet adaptation on METR-LA, combined input outages and delayed feedback yielded an interval-score improvement of 0.355 m/s over rolling calibration (95% paired interval 0.192–0.551) and coverage of 0.8985. Fresh simulation validation under delayed feedback met the rule; permanent-feedback loss missed the 0.88 coverage floor at 0.8793. On PEMS-BAY with permanent-feedback loss, rolling calibration instead performed better by 0.119 m/s. Component removals worsened interval score in 13 comparisons, improved it in four and were inconclusive in 25, based on exploratory paired intervals. These findings support condition-dependent calibration benefits and do not establish universal superiority or correction of unobserved outcomes.

Key words: Traffic forecasting; Prediction intervals; Sensor outages; Delayed feedback; Uncertainty calibration; Reproducibility

1 Introduction

Traffic forecasting is increasingly framed as a prediction problem on a network, but an uncertainty service also depends on a reporting process. A speed measurement has a physical timestamp, a delivery timestamp and a role in subsequent decisions. If a detector gateway stops reporting, the forecasting model loses recent inputs. The calibration module also loses the realised outcomes needed to assess forecasts already issued. Treating these two losses as interchangeable hides an important distinction: a predictor may continue producing plausible values while the residual evidence supporting its intervals becomes stale. The uncertainty problem therefore concerns information availability as well as traffic dynamics. The proposed feedback-aware signed-residual calibrator is named FAR-Cal, and its Graph WaveNet forecasting adaptation is denoted FAR-GW.

The practical motivation is straightforward [1, 2]. An operator receiving a forecast needs to know how much confidence to place in it during degraded sensing. A narrow interval based on a healthy historical period can be misleading after a correlated outage, even when its point forecast changes little. Conversely, widening every interval indiscriminately can make the service uninformative. A useful evaluation must consider coverage together with interval width and the penalty for missed outcomes. It must also retain difficult targets in the scoring population when their reports are unavailable to calibration. Otherwise, removing the very outcomes most affected by failure can artificially improve reported performance.

Delayed feedback presents a second difficulty. A backlog released after restoration may contain many residuals, yet describe conditions several hours earlier. Delivery recency is then a poor proxy for relevance. Permanent loss creates a different information deficit because the missing residuals never enter the live archive. A method that succeeds with delayed packets need not succeed when outcomes disappear. These mechanisms motivate retaining target timestamps, issuing immutable intervals and distinguishing input release from feedback release. The proposed replay follows this ordering explicitly rather than giving the calibrator access to evaluator truth or retrospectively revising uncertainty after the outcome becomes known.

Spatial information offers a possible response, but creates its own risks. Nearby sensors can supply residual evidence when a local detector is silent. Their errors can also be correlated, and their speed regimes need not match the affected location. Context matching and temporal weighting may improve relevance while reducing effective support. An historical archive can stabilise sparse updates, but can preserve obsolete residual behaviour. The methodological question is consequently not whether every available record should be pooled. It is how the usefulness of live, neighbouring and historical evidence changes under a specified reporting process, and how those choices affect interval quality on held-out data.

Traffic predictors provide several ways to represent network dependence. Graph WaveNet and the diffusion convolutional recurrent neural network (DCRNN) use graph-based temporal modelling, whereas the Spatial-Temporal Identity (STID) model and the spatiotemporal adaptive-embedding Transformer (STAEformer) use identity information and adaptive embeddings [3–6]. The present study does not seek to replace

these forecasting architectures. It holds each fitted predictor fixed and compares calibrators on identical forecasts, scales and evaluation masks. This separation makes a paired score difference interpretable as a calibration effect conditional on the backbone. Repeating the comparison across backbones then tests transfer without confusing a stronger point predictor with a better use of outcome feedback.

Conformal prediction (CP) and adaptive conformal inference (ACI) offer foundations, but their assumptions must be respected. ACI adjusts uncertainty from sequential coverage errors, and multistep extensions account for the wait until a forecast target occurs [7, 8]. Published work also studies online distribution shifts and multivariate multistep forecasting [9, 10]. Accordingly, feedback-aware uncertainty is not claimed as an unexplored general topic. Our focus is a traffic replay with independently impaired input and outcome channels, correlated outages, permanent loss and auditable historical fallback. The proposed weighted signed-residual method is evaluated empirically; source theorems are not transferred to its modified reporting protocol.

The research has three connected objectives. First, it establishes an information boundary that prevents unavailable outcomes from entering calibration while allowing originally valid targets to remain assessable by the evaluator. Second, it evaluates whether asymmetric residual pooling improves the registered interval-score endpoint without falling below a prespecified coverage floor. Third, it investigates whether the result survives changes in network, forecasting backbone, reporting severity and individual components. Each objective requires a different comparison. Fresh simulation episodes assess the locked candidate; matched-rate controls examine an alternative outage geometry; component removals assess mechanisms while retaining the same point and scale caches.

The empirical design combines METR-LA, PEMS-BAY and controlled Simulation of Urban MObility (SUMO) simulations. The principal reporting conditions are C3 (input outages with delayed and backlogged feedback) and C4 (input outages with permanent feedback loss). Real-data inference uses chronological partitions and fault-focused hourly origins, while simulation permits full episode evaluation under separately specified physical and reporting events. The study includes core conditions, independent controls, one-factor stress and seven complete component ablations. The breadth is useful only if scope differences remain visible. An hourly network audit is not a complete five-minute deployment trial, and controlled simulation is not proof of field effectiveness. The manuscript therefore reports sampling rules, adaptation choices, uncertainty intervals and adverse findings alongside favourable results, rather than presenting a single aggregated ranking.

The main contribution is an executable causal benchmark coupled with a feedback-aware calibration design and a transparent account of its conditional performance. The final asymmetric method passes twelve of sixteen real-data backbone cells, but this aggregate conceals important differences between networks. Full ablations also reveal components whose removal improves score. Such findings constrain both mechanism claims and practical recommendations. No universal superiority, arbitrary-missingness coverage guarantee or identifiable correction for missing-not-at-random (MNAR) outcomes is asserted. Potential societal benefits motivate the work, while actual impacts

on traffic control, safety, travel time or equity remain outside the demonstrated outcomes.

A reproducible comparison also needs recoverable provenance. Saved schedules, forecast identifiers, configuration hashes and originally valid masks allow readers to distinguish a statistical result from an interrupted runner or a reporting artifact. This accountability is also particularly important when evaluations are resumed, because existing verified stages remain evidence rather than being silently replaced.

The remainder of the paper follows the dependency order of the study. Section 2 positions forecasting, missing-data reconstruction and sequential calibration. Section 3 describes datasets and partitions. Section 4 explains the information flow. Section 5 formalises the model, and Section 6 details its algorithm. Section 7 defines comparisons and inference. Section 8 presents results, Section 9 examines ablations, and Section 10 discusses interpretation and limitations before Section 11 concludes.

2 Related work

2.1 Traffic forecasting backbones and the role of calibration

Journal research places these algorithms within a longer forecasting history. Williams and Hoel [11] model traffic flow as a seasonal ARIMA process, showing how periodic structure informs statistical prediction. Lv et al. [12] develop a deep learning traffic predictor using stacked autoencoders, whereas Polson and Sokolov [13] examine nonlinear short-term forecasting across changing traffic regimes. These studies concern predictive structure rather than our outcome-release intervention. Their findings motivate diverse residual-generating backbones without establishing the efficacy of the present calibrator. None is inserted as a numerical baseline using source-paper scores; historical approaches would require fitting and replay under the same partitions and reporting schedules.

Graph WaveNet combines temporal modelling with learned spatial dependence [3]. Its relevance here is the separation between a forecasting backbone and the uncertainty layer attached to its outputs. A graph predictor can exploit relationships between sensors, yet this does not determine how prediction errors should be used when their reports arrive late. The FAR-GW adapter adds causal validity and age channels and a noncrossing quantile head. These are documented study modifications rather than claims about the original implementation. Table 1 distinguishes the forecasting references from the uncertainty mechanisms evaluated on their frozen outputs.

Table 1 Forecasting and input-reconstruction literature: role in this study.

Reference	Representation or objective	Numerical study role
Graph WaveNet [3]	Graph supports and temporal convolutions	FAR-GW causal quantile adapter
DCRNN [4]	Diffusion recurrent forecasting	Causal quantile adapter; data provenance
STID [5]	Spatial and temporal identity	Causal quantile adapter
STAEformer [6]	Adaptive embeddings and Transformer	Independent paper-based adapter
Graph Recurrent Imputation Network (GRIN) [14]	Graph-based missing-value reconstruction	Literature context; not evaluated

Representation choices do not determine how missing outcome feedback should be handled. Source-paper performance values are not used as study results.

DCRNN represents spatial propagation through diffusion operations within a recurrent forecasting architecture [4]. It supplies a contrasting temporal backbone and the provenance of the real benchmark packages. In this study, its multihorizon quantile adaptation uses the same causal channels and reporting schedules as the other predictors. The purpose of including it is to examine whether calibration effects persist under a different residual-generating model. Source-paper scores cannot serve as the comparison because dataset partitions, valid-target masks and fault processes differ. Only predictions generated under the common study protocol enter the paired backbone analysis.

STID provides a relatively simple forecasting design built around spatial and temporal identity representations [5]. STAEformer instead uses adaptive spatiotemporal embeddings with a Transformer [6]. These references demonstrate that useful traffic representations need not rely on the same graph-convolution construction. Their inclusion broadens the residual shapes encountered by the calibrator. It does not make the present study an exhaustive forecasting benchmark: only four fitted architectures are evaluated, and each has a documented adaptation. The distinction between model implementation, training protocol and calibration evaluation remains essential when interpreting transfer across them.

2.2 Missing observations and uncertainty calibration

Missing-input reconstruction is a related but distinct objective. GRIN uses graph message passing to reconstruct incomplete multivariate series [14]. An imputation method addresses what value the forecasting model can consume when a channel is absent; a feedback-aware calibrator addresses which forecast errors have actually become available. The two can interact, because input reconstruction changes the subsequent residual distribution, but they cannot be substituted for each other. Our causal last-observation and training-median fills are a deliberately explicit input policy.

GRIN is discussed as relevant literature, not included as an unevaluated numerical comparator.

Conformal prediction separates an underlying predictor from a calibration procedure, as explained by Angelopoulos and Bates [15] and formalised for regression by Lei et al. [16]. A central distinction is between an empirical coverage estimate and a guarantee derived under a specified sampling assumption. Traffic observations are temporally dependent, and artificial outages further change which outcomes reach the service. Consequently, a calibration construction cannot acquire distribution-free validity merely because it uses residual quantiles. We retain the terminology of calibration while identifying the proposed rule as an empirical weighted signed-residual method. Its nominal interval level, observed coverage and registered decision tolerance are reported separately throughout the results.

Conformalized quantile regression (CQR) combines quantile modelling with conformal calibration to accommodate heteroscedastic responses [17]. This is relevant to the use of a predictive centre and scale, because residual magnitudes need not be comparable across locations or traffic regimes without normalisation. Our quantile head supplies that scale but is not itself treated as a calibrated uncertainty guarantee. The final interval comes from received signed residuals and a fallback archive. The mathematical formulation therefore distinguishes predictor quantiles from calibration quantiles, making clear which quantities are fitted before test replay and which change after causal feedback release.

Barber et al. [18] analyse conformal prediction beyond exchangeability, including weighted quantiles and distribution drift. Their work provides a useful caution for interpreting relevance weights. Emphasising recent or contextually similar observations can address changing distributions, but the validity of a specific weighting scheme depends on its own assumptions and construction. The exponential target-age and context weights used here are not presented as an instantiation of that paper's complete theorem. They express a testable engineering hypothesis about useful residual evidence. The full ablations are needed precisely because plausible weighting components may help one network and harm another.

2.3 Sequential, multistep and relational inference

ACI updates a control variable from sequential coverage errors [7]. This establishes an important alternative to keeping a calibration archive completely fixed. For our comparison, the ACI adapter is horizon specific and updates once per spatial feedback batch. It cannot update before a packet arrives and receives no update from a permanently lost target. These changes define the actual comparator rather than the original source protocol. A favourable or unfavourable paired result therefore concerns the implemented adaptation under common causal replay, not a refutation or confirmation of the source theorem.

Sequential predictive conformal inference develops adaptive prediction intervals for time series by exploiting sequential residual information [19]. Its relevance is the broader principle that temporal dependence can carry information about future uncertainty. That principle supports examining a residual archive beyond an unordered

static collection. However, relevance of historical residuals and availability of their outcomes remain separate questions. We discuss sequential predictive inference to position the problem, without reporting comparative scores that were never generated. Adding it as an executable comparator would require a separately specified feedback-arrival adaptation and a new registered evaluation.

Multistep adaptive conformal inference accounts for the fact that the outcome of a forecast is unavailable until its target time [8]. This ordinary horizon delay should not be confused with an additional communication backlog. A fifteen-minute forecast can have its outcome physically realised after fifteen minutes yet delivered much later. Our replay records both target and release timestamps and permits permanent loss. The ACI implementation uses the multistep reference as its starting point but changes spatial batching and reporting availability. Table 2 records this distinction without implying that the source method ignores ordinary forecasting delay.

Table 2 Calibration literature and feedback scope.

Reference	Relevant objective	Scope here
CQR [17]	Calibrated quantile regression	Literature; predictor quantiles distinguished from calibrated bounds
Beyond exchangeability [18]	Weighted inference under drift	Literature; no theorem imported
ACI [7]	Sequential coverage adaptation	Horizon-specific spatial-batch adapter
Sequential predictive conformal inference (SPCI) [19]	Sequential residual-based intervals	Literature; not evaluated
Multistep ACI [8]	Ordinary forecast-target delay	Starting point for ACI adaptation
Conformal Relational Prediction (CoRel) [20]	Relational series calibration	Official architecture; FAR-GW only
Incomplete series [21]	Post-training missing-input adaptation	Published 2026; literature comparison
Online inference [9]	Local adaptation under distribution shifts	Journal method; not evaluated
Ensemble conformalized quantile regression (EnCQR) [22]	Ensemble quantile calibration for time series	Transactions method; not evaluated

Entries summarise reviewed objectives, not universal capability claims or a numerical ranking. General CP background is supplied by Angelopoulos and Bates [15].

CoRel uses relationships between correlated time series to calibrate uncertainty on top of a pretrained predictor [20]. This is the nearest directly evaluated relational comparator and prevents an overbroad claim that neighbouring series have not previously

been used for uncertainty estimation. Its study adaptation follows the official architecture while changing partitions, multihorizon targets and residual availability to match our protocol. CoRel is evaluated only with FAR-GW caches. The comparison consequently isolates calibration on a common predictor; it neither reproduces every source setting nor supports conclusions about CoRel on untested forecasting backbones.

2.4 Recent journal studies and incomplete observations

Chen et al. [21] study post-training adaptive conformal prediction for incomplete time series. The previously audited method uses imputation and missing-pattern-conditioned calibration, followed by adaptation when the test outcome is observed. Its substantial overlap with our motivation is acknowledged: incomplete histories can require uncertainty adjustments beyond ordinary clean-input calibration. Our different evaluation target concerns the arrival or permanent suppression of outcome feedback, represented separately from predictor-input availability. The distinction is limited to the reviewed protocol; it is not a claim that the entire incomplete-data literature lacks delayed reporting or that our method identifies unseen outcomes.

Gibbs and Candès [9] extend online conformal inference with adaptation designed to respond to changing distributions over local time intervals. This journal contribution is distinct from the original ACI reference and highlights the potential cost of over-weighting historical data. Jensen et al. [22] combine ensemble learning, quantile regression and conformal calibration for probabilistic time-series forecasting. Their treatment of heterogeneous predictive uncertainty complements the motivation for scale normalisation. Neither journal method is numerically evaluated here. Their published guarantees and empirical findings cannot be transferred to our selectively received residuals merely by sharing a calibration vocabulary.

Schlembach et al. [10] study conformal multistep multivariate forecasting, combining multiple targets with methods for nonexchangeable observations. This expands the relevant literature beyond single-target intervals but does not make horizon delay identical to a communication outage. The survey by Zhou et al. [23] organises conformal prediction around data characteristics, including dynamic settings. Together, these journal contributions motivate specifying the data and reporting assumptions before interpreting uncertainty. Our contribution remains the executable traffic protocol and its measured conditional results. It is not a claim that adaptive inference, multivariate calibration or uncertainty for incomplete observations originated in this study.

2.5 Positioning and evidence boundary

Table 1 summarises representation choices and implementation scope; Table 2 summarises uncertainty objectives, feedback assumptions and whether a method was numerically evaluated here. These tables are literature maps, not score tables. Numerical comparisons appear only in the results and share forecasts and masks. SUMO supplies controlled simulation for examining reporting and physical events separately [24]. Combining these resources permits a focused benchmark contribution, while the recent literature rules out a broad first-of-its-kind claim for adaptive, relational

or delayed-feedback inference. The remaining research question is conditional traffic interval quality under the precisely specified causal protocol.

Literature relevance does not imply experimental reproduction. Numerical claims require common frozen predictions, causal schedules and paired outcomes. Unevaluated methods remain explicit future comparisons, while component removals test the proposed mechanism within this protocol.

3 Datasets and preprocessing

3.1 METR-LA and PEMS-BAY

METR-LA comprises 207 sensors and 34,272 five-minute timestamp rows from 1 March to 27 June 2012. PEMS-BAY comprises 325 sensors and 52,116 original rows from 1 January to 30 June 2017. These statements describe the acquired benchmark files, whose provenance is recorded against DCRNN [4]; the graph and speed-file hashes are retained separately. PEMS-BAY has a 65-minute timestamp gap. Reindexing inserts twelve wholly invalid five-minute rows, giving 52,128 prepared rows. METR-LA requires no inserted rows. Table 3 compares the dimensions, periods, missingness convention and sampling policy. Fig. 1 shows training-partition speed distributions and original validity, making the empirical input populations visible without treating distributional differences as a causal explanation for transfer results.

Table 3 Dataset dimensions, periods and evaluation sampling.

Data	Sensors	Rows	Acquired/prepared period	Test sampling
METR-LA	207	34,272	1 Mar–27 Jun 2012	Hourly fault audit
PEMS-BAY	325	52,128	1 Jan–30 Jun 2017	Hourly fault audit
SUMO original	20	11,520	120 eight-hour episodes	Full episode audit
SUMO fresh C3	20	3,840	40 fresh episodes	Full episode audit

All rows use five-minute bins. PEMS-BAY has 52,116 raw rows plus twelve inserted wholly invalid rows. Real zero speeds follow the benchmark missingness convention; physical SUMO zeros remain valid.

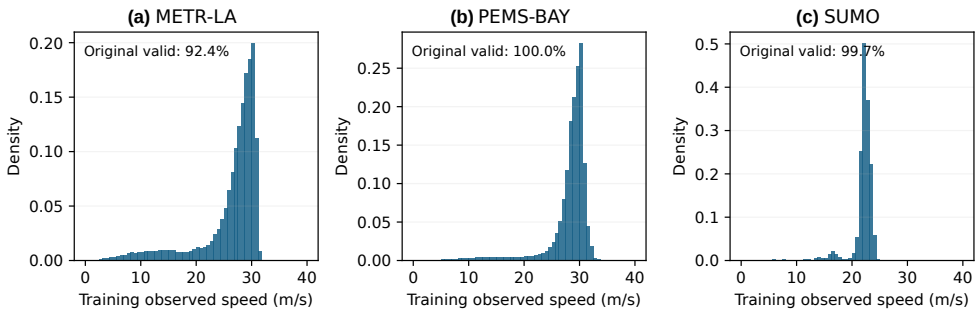


Fig. 1 Training-partition observed-speed distributions before forecast-window construction. Only originally valid prepared measurements are plotted; labels report original valid fractions. SUMO includes its prepared training-partition measurements. The distributions are descriptive, not an explanation of calibration transfer. Panels (a)–(c) show METR-LA, PEMS-BAY and SUMO, respectively.

The benchmark convention treats zero-valued real speeds as missing, following the audited source mask. It should not be interpreted as a physical assertion that vehicles never stop. Originally invalid entries cannot become training labels, calibration residuals or evaluation targets after filling. SUMO instead retains physical zero speeds as valid and masks empty detector intervals. Table 4 reports original validity and observed-speed summaries computed directly from prepared arrays. These are descriptive data statistics, not estimates from the artificially censored reporting stream.

Table 4 Descriptive prepared-data validity and observed speeds.

Data	Valid (%)	Median	5th percentile	95th percentile
METR-LA	91.89	28.26	11.46	30.80
PEMS-BAY	99.97	29.19	17.88	31.16
SUMO	99.73	22.31	16.65	23.66

Speed summaries are in $\mathrm{m\,s^{-1}}$ across originally valid prepared measurements before reporting interventions. They are not distributional estimates for the selectively received feedback stream.

3.2 Chronological separation and causal features

Real-data boundaries are selected on the original timestamps before reindexing and window construction. The split proportions are 60% training, 10% tuning, 10% initial calibration and 20% testing. METR-LA boundaries are 11 May 2012 09:35, 23 May 07:10 and 4 June 04:45; PEMS-BAY boundaries are 19 April 2017 14:45, 7 May 17:05 and 25 May 19:20. Table 5 provides these timestamps and the simulation episode allocations. A forecast window belongs to a partition only when all its targets remain inside that partition. Interpolation across a future boundary is prohibited.

Table 5 Chronological boundaries and episode separation.

Data	Train/tune boundary	Tune/calibration boundary	Calibration/test boundary
METR-LA	11 May 2012 09:35	23 May 2012 07:10	4 Jun 2012 04:45
PEMS-BAY	19 Apr 2017 14:45	7 May 2017 17:05	25 May 2017 19:20
SUMO original	60 training episodes	15 tuning episodes	15 calibration; 30 test episodes
SUMO fresh C3	No predictor retraining	Candidate locked before generation	40 new validation episodes

Real proportions are 60/10/10/20 percent. Targets must remain inside their assigned partition. Timestamp boundaries precede PEMS-BAY gap reindexing.

Every model receives twelve historical bins, corresponding to 60 minutes, and forecasts twelve future bins. Reported endpoints use horizons three, six and twelve, corresponding to 15, 30 and 60 minutes. Missing input values use the latest available past observation, with the training sensor median as fallback when no prior valid measurement exists. Filling changes the predictor input, not original validity. Seven channels comprise standardised filled speed, validity, capped observation age divided by 288, and sine/cosine representations of time of day and day of week. Scaling, sensor medians, graph processing and context normalisation use training data only. During replay these features are reconstructed from released packets rather than from a fully observed test array.

3.3 Controlled SUMO datasets

SUMO/TraCI 1.27.1 is used for a controlled 10-km, two-lane mainline with 20 detector stations [24]. The simulator steps every second; detector values are aggregated every 300 seconds. Each episode lasts eight hours, with a one-hour warm-up. The original prepared corpus contains 120 episodes and 11,520 timestamp rows. Sixty episodes train the predictor, fifteen tune it, fifteen initialise calibration and thirty form the held-out test set. Causal state resets at episode boundaries, and synthetic inter-episode gaps are not interpreted as observed traffic.

Four registered physical families distinguish baseline demand (P0), increased demand (P1), a temporary speed restriction (P2) and their combination (P3). The demand generator mixes 90% passenger vehicles and 10% trucks. For peak families, the sampled demand multiplier is 1.4–1.8. Restriction families temporarily lower the speed on mainline edge m15 to 8.33 $\mathrm{m\,s^{-1}}$ for a sampled 1,200–2,400 seconds, then restore 27.78 $\mathrm{m\,s^{-1}}$. These are simulated physical events, separate from packet-release faults. They enable repeatable stress without claiming to reproduce every real congestion mechanism.

Further episode sets are generated for independent replication, powered confirmation and final asymmetric validation. The latter contains forty fresh episodes, balanced

across P0–P3, with generation seeds 70001–70040. The locked asymmetric C3 candidate is evaluated on this fresh set; C4 transfer uses the thirty original held-out episodes. The separation is recorded explicitly so favourable fresh validation is not confused with an untouched C4 experiment. Prepared simulation arrays, generation settings and source hashes accompany the public release; raw real benchmark packages require separate acquisition because their exact redistribution rights remain unresolved.

3.4 Observed targets and audit sampling

The real evaluation is a fault-focused audit: origins are selected hourly and affected sensors are followed through twelve post-event bins. Simulation analyses use full registered episode sets. Both retain originally valid targets even if outcome feedback is delayed or lost. This distinction prevents scoring from favouring a method merely because difficult labels disappeared from its update stream. Table 3 therefore states the evaluation frequency alongside dataset dimensions. The dataset profile describes measurement populations; it does not establish national, seasonal or road-type representativeness, which remains a limitation of the study.

4 System architecture

4.1 Separate physical, reporting and evaluation layers

Fig. 2 depicts the redesigned system in separate lanes with orthogonal routes. The upper lane produces and releases observations; the middle lane forms immutable forecasts; the lower lane matches released feedback and updates calibration. An evaluator compartment remains outside these service lanes. This organisation distinguishes data movement from statistical operations and prevents arrows from passing through module labels. Table 6 specifies the inputs, outputs and information restrictions of each module, complementing the diagram rather than relying on visual proximity to imply an interface.

Table 6 Architecture interfaces and information boundary.

Module	Consumes	Produces / restriction
Input store	Released speed, validity and measurement time	Past-only seven-channel history; no future packets
Frozen predictor	Available history and fitted parameters	Median and floored scale; no test-time training
Issue ledger	Point, scale, context and interval	Immutable sensor/target/horizon records
Feedback matcher	Released valid outcomes and stored records	Residual once per forecast–horizon key
Calibration kernel	Received live and valid frozen archives	Weighted signed-tail bounds; no hidden truth
Evaluator	Original validity, truth and issued bounds	Scores; no return edge to service

The diagram separates service data flow from evaluator knowledge. Recorded software/artifact checks establish replay properties, not operational latency.

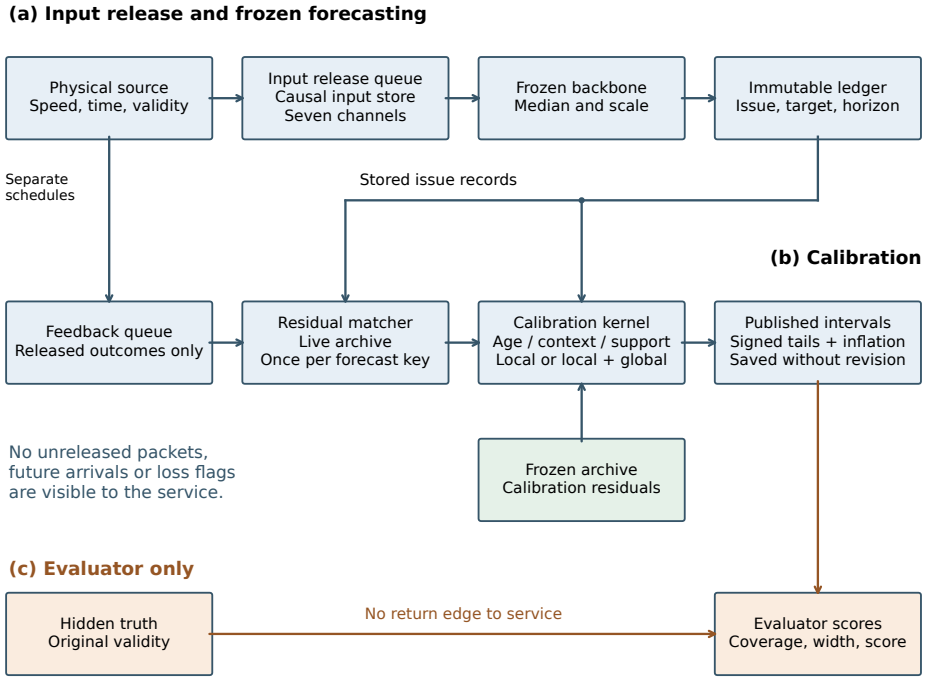


Fig. 2 Causal architecture with separate input, feedback and evaluator lanes. Stored issue records feed both residual matching and the calibration query; the junction is intentional. Hidden truth feeds scoring only. All module routes avoid text and preserve the service information boundary.

The physical source supplies speed, timestamp, sensor identity and original validity. In simulation it also supplies episode identity. An evaluator-side fault generator produces separate input and feedback release schedules before replay. Its permanent-loss markers and future schedule entries are hidden from the service. The input release queue updates a causal store, which constructs the seven-channel history described in Section 3. A separate feedback queue releases outcomes that can be matched to existing forecasts. The evaluator retains the uncensored truth and original mask exclusively for scoring.

4.2 Frozen prediction and immutable issuance

The forecasting lane contains a fitted backbone and noncrossing 0.05/0.50/0.95 head. For FAR-GW, gated dilated temporal convolutions interact with fixed and learned graph supports. The other backbone adapters replace this representation module while preserving the input interface and quantile output format. The median is the point forecast, and half the outer-quantile spread supplies a floored normalisation scale. Parameters remain frozen throughout calibration and test replay. Only the calibration state changes when eligible outcomes arrive.

Before publication of an interval, the issuance service creates a record containing sensor, issue time, target time, horizon, point, scale, context and interval bounds. The issued record is immutable. This ledger is necessary because a delayed target must be compared with the forecast actually made for it, not a later forecast generated after restoration. Model and calibrator versions and source manifests support auditing. Outcomes shared by several forecast horizons are legitimate, but duplicate ingestion of the same forecast–horizon residual is prohibited.

4.3 Feedback matching, archives and calibration

When a valid feedback packet arrives, the matcher retrieves its eligible forecast records and computes residuals from their stored centres and scales. The live archive retains target timestamps and issue contexts. A frozen archive from the calibration partition supplies fallback evidence. Target-time expiry is applied before querying live pools, so an old report cannot become recent merely because it was delivered in a backlog. No residual is inferred for a permanently missing report.

The calibration kernel queries a local graph group, applies target-age and context weights, measures timestamp-block support and computes a freshness gap. The real-data path uses capped local pools; the SUMO path also includes a global pool. These paths are explicit in the diagram and algorithm because their sampling and pooling estimands differ. The mixture shifts mass towards frozen evidence when live support is stale or insufficient. Signed-tail quantiles, bounded inflation and nonnegative clipping then produce the final interval. The output is stored before evaluator scoring can occur.

4.4 Execution and reproducibility boundary

Replay advances in five-minute bins: release packets, update causal inputs, ingest feedback, expire live records, query the frozen predictor and issue intervals. Evaluator-only scoring compares saved intervals with originally valid outcomes. Fig. 2 shows

scoring as a terminal branch without a return edge to the service. This ordering is the core protection against information leakage, not a claim that every proposed weight is statistically optimal.

The implementation checks source manifests and chunk hashes before accepting completed evaluations. Resumed runners may revisit artifact manifests without discarding verified work. This is an execution property rather than an additional statistical replicate. Similarly, the gateway smoke tests establish limited replay-equivalence behaviour, not production latency, network reliability or operational readiness. Table 6 makes those boundaries reviewable: the source and evaluator may know truth, but the predictor and calibrator receive only the information already released to their respective channels.

5 Mathematical modelling

5.1 Causal information and forecast targets

Table 7 defines the principal symbols and units; Tables 8 and 9 complete the nomenclature for calibration, prediction, training and replay. The model distinguishes physical validity from channel availability; the former controls scoring and the latter controls service access.

Table 7 Nomenclature: principal mathematical symbols.

Symbol	Meaning	Units
$G = (V, E)$	Sensor graph	–
t, u, h	Issue bin, target bin, forecast horizon	Five-minute bins
a^x, a^f	Input and feedback release times	Bins or infinity
y, m, s	Target speed, point centre, scale	$\mathrm{m\,s^{-1}}$
r	Signed normalised forecast residual	Dimensionless
c, z	Issue context and missing fraction	Dimensionless
R_h, A_h	Live and frozen residual archives	Records
B, τ, d	Retention, age and stale-decay lengths	Bins
G_i, C	Local group and group-cap multiplier	Sensors / records
b, n_0	Context bandwidth and support reference	Dimensionless
n_{eff}, g	Timestamp-block support and freshness gap	Blocks / bins
η, λ, κ	Archive mixture, local mixture, inflation	Dimensionless
L, U, IS	Interval bounds and interval score	$\mathrm{m\,s^{-1}}$
α	Nominal miscoverage	Probability

International System of Units (SI) conversion uses exactly 0.44704 metres per second per mile per hour (mph). Dimensionless mixture weights and coverage remain unchanged.

Table 8 Nomenclature: residual records and calibration distributions.

Symbol	Meaning
i, k	Sensor index and retained residual-record index
u_k, c_k	Target timestamp and stored issue context of record k
$w_k, \widetilde{w}_k$	Unnormalised and normalised relevance weights
W_u	Sum of relevance weights for records with target timestamp u
P, Q	Local and global eligible live-residual pools
F_P, F_Q	Empirical weighted distributions of the local and global live pools
F_A	Frozen-archive distribution used by the stated implementation path
F	Mixture distribution queried for signed-tail quantiles
$Q_p(F), p$	Empirical inverse-distribution quantile and its probability level

Table 9 Nomenclature: predictor, training and replay quantities.

Symbol	Meaning
$x_{i,u}, o^x_{i,u}(t)$	Input measurement and its valid-and-released indicator at issue time t
$\mathcal{I}_t, \mathcal{L}$	Service information set and immutable forecast ledger
f_θ, θ	Frozen forecasting function and its fitted parameters
$\widehat{q}_{p,i,t,h}, s_0$	Predictor quantile and positive scale floor, in speed units
X_t	Twelve-bin, seven-channel causal predictor-input tensor
$A_{\text{adaptive}}, E_1, E_2$	Learned graph support and trainable node factors
$W_f, W_g, *, \odot$	Temporal filter/gate parameters, convolution and elementwise product
$\sigma, \text{ReLU}, \text{softplus}$	Logistic sigmoid, rectified linear unit and softplus activations
a, b, c	Local raw head coordinates; distinct from context-bandwidth b
$\mathcal{M}, \mathcal{P}$	Valid training-target index set and fitted quantile-level set
$e, \rho_p(e), \mathcal{J}(\theta)$	Prediction error, pinball loss and masked training objective
$\mathbf{1}(\cdot)$	Indicator of whether the stated event occurs

Let $G = (V, E)$ represent the sensor graph, $y_{i,u}$ the speed at sensor i and target bin u, and t an issue bin. The reporting channel assigns input arrival $a^x_{i,u}$ and feedback arrival $a^f_{i,u}$. Permanent loss is represented by an infinite arrival time. A value may enter the predictor only when $a^x_{i,u} \le t$ and may update calibration only when $a^f_{i,u} \le t$. The service information set is

$$\mathcal{I}_t = \{x_{i,u} : a^x_{i,u} \le t\} \cup \{y_{i,u} : a^f_{i,u} \le t\} \cup \{\text{previously issued records}\}. \tag{1}$$

The predictor f_θ is frozen and depends only on $\mathcal{I}_t$. For horizon h, define its centre $m_{i,t,h}$ and scale

$$m_{i,t,h} = \widehat{q}_{0.50,i,t,h}, \qquad s_{i,t,h} = \max\{(\widehat{q}_{0.95,i,t,h} - \widehat{q}_{0.05,i,t,h})/2, s_0\}, \tag{2}$$

where $s_0 = 0.44704\,\mathrm{m\,s^{-1}}$, equivalent to the implementation's one-mile-per-hour floor. All reported dimensional scores are converted to SI units using the exact factor $1\,\mathrm{mph} = 0.44704\,\mathrm{m\,s^{-1}}$; this rescaling does not change coverage or comparison signs.

5.2 Residual eligibility and relevance weighting

A signed residual becomes eligible only after its target report arrives:

$$r_{i,t,h} = (y_{i,t+h} - m_{i,t,h})/s_{i,t,h}, \qquad a^f_{i,t+h} \le t', \tag{3}$$

where t' is the current calibration time. The original issue-time centre and scale are used. For each horizon, the live archive $R_h(t')$ retains received residuals whose targets are at least $t' - B$, with $B = 2016$ five-minute bins. Target age, rather than delivery age, governs this eligibility.

Write $c_{i,t}$ for a two-dimensional issue context: neighbourhood mean availability and the logarithm of one plus neighbourhood mean observation age, each standardised using training histories. For a pool P, a record k with target u_k receives unnormalised weight

$$w_k(t', c) = \exp\left[-\frac{t' - u_k}{\tau} - \frac{\|c_k - c\|_2^2}{2b^2}\right], \qquad \widetilde{w}_k = w_k / \sum_{j \in P} w_j. \tag{4}$$

The locked settings are $\tau = 288$ bins and $b = 2$. Grouping weights by target timestamp avoids counting correlated spatial reports at the same time as independent evidence. With $W_u = \sum_{k:u_k=u} w_k$, the support measure is

$$n_{\mathrm{eff}}(P) = \frac{(\sum_u W_u)^2}{\sum_u W_u^2}. \tag{5}$$

This is a heuristic effective-support statistic, not a claim that dependent samples become independent.

5.3 Archive mixtures and stale support

The local pool contains sensor i and up to two graph neighbours. Let g be the non-negative gap between the current bin and the latest local target, using archive targets as fallback when no live record is available. With reference support $n_0 = 50$ and decay $d = 288$ bins, the real-data local implementation uses

$$\eta = \begin{cases} \min(1, n_{\mathrm{eff}}(P)/n_0)\exp(-g/d), & P \neq \varnothing, \\ 0, & P = \varnothing. \end{cases} \qquad F = \eta F_P + (1-\eta)F_A, \tag{6}$$

where F_A is the uniformly weighted frozen archive distribution for the local group, with archive fallback. The local query retains at most $256|G_i|$ records in archive order, where G_i is the sensor group; this is a group limit rather than an equal per-sensor allocation.

The uncapped SUMO implementation also includes a global live pool Q. Its live distribution is $\lambda F_P + (1-\lambda)F_Q$, where $\lambda = n_{\mathrm{eff}}(P)/(n_{\mathrm{eff}}(P) + n_0)$. Equation (6)

then uses global support for η and a sensor-specific archive, falling back to the full archive if fewer than 50 sensor records are available. Both paths implement the same feedback-aware principle, but their pooling estimands and sampling rates differ and are reported separately.

5.4 Signed-tail interval construction

Let z denote issue-time missingness, recovered from the stored availability context. Bounded inflation is

$$\kappa = \min\{3, 1 + 0.05 \min(g/d, 2) + 0.05z\}. \tag{7}$$

For empirical weighted quantile $Q_p(F) = \inf\{q : F(q) \geq p\}$ and nominal miscoverage α, the locked fully asymmetric interval is

$$L_{i,t,h} = \max\{0, m_{i,t,h} + s_{i,t,h}\kappa Q_{\alpha/2}(F)\}, \tag{8}$$

$$U_{i,t,h} = \max\{L_{i,t,h}, m_{i,t,h} + s_{i,t,h}\kappa Q_{1-\alpha/2}(F)\}. \tag{9}$$

The candidate uses asymmetric blend one, so no symmetric rolling-radius component remains. Multiplication of signed tails by κ need not strictly enlarge both sides when both tail quantiles have the same sign; the equations specify the implemented transformation rather than asserting monotone widening in every case.

5.5 Predictor input, training objective and noncrossing head

Let $o^x_{i,u}(t)$ indicate that the value at timestamp u is valid and released to the input store by issue time t. Its age feature is the time since the most recent available observation, capped at one day and divided by 288. Together with training-standardised past-filled speed and four calendar channels, this produces $X_t \in \mathbb{R}^{12\times|V|\times 7}$. The validity channel records actual availability even when filling provides a numerical speed. Thus numerical completeness of X_t does not imply that the detector is currently reporting.

For FAR-GW, the learned graph support is

$$A_{\text{adaptive}} = \text{softmax}_{\text{row}}\{\text{ReLU}(E_1E_2)\}, \tag{10}$$

where E_1, E_2 are trainable node factors. Gated temporal features have the form $\tanh(W_f * X) \odot \sigma(W_g * X)$, with dilated temporal convolutions and graph diffusion projections. These backbone operations generate three raw head outputs a, b, c, converted to

$$\widehat{q}_{.05} = b - \text{softplus}(a), \quad \widehat{q}_{.50} = b, \quad \widehat{q}_{.95} = b + \text{softplus}(c). \tag{11}$$

Equation (11) prevents quantile crossing by construction; it does not guarantee empirical coverage. Other backbone adapters preserve the same noncrossing output interface.

With valid training target index set $\mathcal{M}$ and $\mathcal{P} = \{.05, .50, .95\}$, training minimises masked pinball loss [25]

$$\mathcal{J}(\theta) = \frac{1}{3|\mathcal{M}|} \sum_{p \in \mathcal{P}} \sum_{(i,t,h) \in \mathcal{M}} \rho_p(y_{i,t+h} - \widehat{q}_{p,i,t,h}), \quad \rho_p(e) = \max\{pe, (p-1)e\}. \tag{12}$$

The implementation evaluates this loss in standardised training units. Empty-validity batches are skipped, and tuning loss selects the checkpoint without test outcomes. The fitted head supplies the centre and scale in Eq. (2); received residuals, rather than further predictor training, update the uncertainty layer during replay.

5.6 Identifiability and computational assumptions

The empirical mixture requires a nonempty valid initial archive. Its normalised weights are computed after subtracting the largest log weight for numerical stability, which leaves relative weights unchanged. Zero-mass entries are excluded before quantile lookup. Timestamp-block support reduces the tendency to treat simultaneous neighbouring records as separate independent observations, but is not an independence proof. No finite-sample correction beyond the stated empirical inverse-distribution quantile is applied.

Permanent feedback loss removes information about realised errors. If loss depends on unseen speed, the received residual distribution need not equal the full target distribution. Neither spatial pooling nor historical fallback identifies that difference without additional assumptions. The MNAR stress therefore measures sensitivity to the specified low-speed suppression process. Equations (6) and (9) describe the executable heuristic; they do not establish arbitrary-missingness coverage or recover outcomes that the service never received.

6 Proposed algorithms

6.1 Causal replay and residual ingestion

Algorithm 1 specifies the state transitions needed to reproduce the study. Its inputs are frozen model parameters, partition-safe calibration records, graph groups and two evaluator-owned release schedules. Its outputs are immutable interval records and evaluator metrics. The schedules determine packet availability, but their unreleased entries are never exposed to the model. Steps involving truth are confined to the evaluator, while residual ingestion requires an actually released, valid outcome. The procedure applies separately at each episode boundary in simulation.

Algorithm 1: causal replay with immutable forecasts

1. **Initialise:** freeze model and calibration parameters; load graph groups, training statistics and calibration archive A_h for each reported horizon h.
2. Initialise causal input store, live buffers R_h, issue ledger $\mathcal{L}$ and an ingestion-key set. Reset causal episode state when required.
3. **For each issue bin t:** obtain only input packets released by t; retain measurement timestamps and validity, and update the causal store.
4. Obtain only feedback packets released by t. For each valid packet (i, u, y), retrieve ledger entries whose sensor is i and target is u.
5. For each matching entry (i, v, h), check $u = v + h$ and its unique ingestion key. Skip entries already ingested; otherwise compute $(y - m_{i,v,h})/s_{i,v,h}$.
6. Append the signed residual, target u, arrival t, sensor and stored issue context to R_h. Mark the key ingested. Never revise the saved interval.
7. Expire live records with target $u < t - B$. Retain A_h unchanged; delivery time does not reset target age.
8. Build the available twelve-bin history with past-only filling and training fallbacks. Compute issue context and missing fraction without hidden outcomes.
9. Query frozen f_θ to obtain median $m_{i,t,h}$ and floored scale $s_{i,t,h}$ for every sensor and reported horizon.
10. Call Algorithm 2 using R_h, A_h, current context and graph group. Store $(i, t, t+h, h, m, s, c, L, U)$ in $\mathcal{L}$ and publish $[L, U]$.
11. **Evaluator only:** score the saved interval against originally valid truth; missing feedback does not remove the target from evaluation.
12. At completion, aggregate the prespecified horizons and paired blocks; save metrics, source hashes and inference units. Validate artifacts before reuse.

The ingestion key is defined per issued forecast and horizon, not merely per measured target. This allows one outcome to assess multiple previously issued horizons without counting the same residual twice. The immutable ledger also prevents a restoration-time forecast from replacing an outage-time forecast during scoring. If no outcome arrives, the loop performs no residual update for that target. These cases explain why release ordering and matching are substantive parts of the method rather than administrative details.

6.2 Weighted asymmetric calibration kernel

Algorithm 2 expands the residual-to-interval calculation. It follows Eqs. (4)–(9) and makes local versus local/global processing explicit. Weighted quantiles are empirical left-continuous inverse-distribution queries; they do not add an unimplemented conformal correction. The frozen fallback ensures that loss of live support does not force an empty query, provided the valid initial archive exists. Inflation is the implemented signed-tail scaling and need not widen both sides when the two selected quantiles have the same sign.

Algorithm 2: asymmetric interval kernel

1. **Inputs:** $i, t, h, m, s, c, z, R_h, A_h$; group G_i; locked parameters B, τ, b, n_0, d, inflation coefficients and archive cap C.
2. Select received live records in G_i. In real-data local mode, retain at most $C|G_i|$ last group records in archive order and apply the same group limit to frozen records.
3. Compute log weights $-(t-u_k)/\tau - \|c_k - c\|^2/(2b^2)$. Subtract their maximum before exponentiation, normalise, and group weights by target timestamp for n_{eff}.
4. Determine freshness gap from the latest local target; use local archive targets if the live group is empty. Apply the documented full-archive fallback when necessary.
5. **Real mode:** set η from local support and gap. Mix local weighted signed residuals with a uniformly weighted frozen group archive.
6. **SUMO mode:** also query and weight the global live pool. Set λ from local support and η from global support; mix local, global and frozen sensor evidence. If fewer than fifty frozen sensor records exist, use the full archive.
7. If no eligible live records exist, set live mass to zero. Compute κ from freshness and issue missingness, bounded by the registered cap.
8. Remove zero-mass entries; sort remaining signed residuals stably. Form cumulative weights and take the first residual whose cumulative mass reaches $\alpha/2$ and $1-\alpha/2$ of total mass.
9. Form $L = \max(0, m + s\kappa q_{\alpha/2})$ and $U = \max(L, m + s\kappa q_{1-\alpha/2})$; return the bounds and query diagnostics.
10. **Ablation mode:** change only the registered component: neighbourhood, context, age weights, stale mixture, inflation, freshness timestamp or frozen-only path. Preserve forecasts, masks and all other settings.

With K retained records, weight formation is linear in K and the direct stable quantile sort costs $O(K \log K)$ per query. These are algorithmic operation counts, not measured throughput. Capped real queries bound the number of local records by $C|G_i|$; they do not guarantee equal allocation of records to each group sensor. Section 9.1 reports the separate descriptive timing experiment. No asymptotic argument is used to claim a speed advantage over the other calibrators.

7 Experimental protocol

7.1 Comparators and adaptation scope

Rolling calibration uses causally received absolute normalised residuals by sensor and horizon. The ACI adaptation maintains a horizon-specific significance controller. At a feedback arrival bin it updates once using the spatial batch's mean interval miss; there is no update before release and none for a permanently lost outcome. This batching and selective outcome suppression differ from the source protocol, so the source coverage guarantee is not inherited [8].

CoRel is evaluated as a documented project adaptation on the FAR-GW prediction caches. Its architecture and relational calibration follow the official implementation, but the partitioning, multihorizon targets and reporting replay follow this study. These results compare calibration on identical frozen forecasts, not the scores reported on the source paper's splits [20]. CoRel is not extrapolated to the other backbones.

FAR-GW adapts Graph WaveNet with the seven causal channels and noncrossing quantile head. STAEformer is an independent paper-based implementation because the audited official repository did not provide a licence file. STID and DCRNN are documented adaptations of their licensed references. Each backbone is trained with seeds 11, 22 and 33; fault seeds are 101, 202 and 303. Table 10 records shared settings and calibrated-method parameters. Per-backbone configurations and source audits accompany the release.

Table 10 Shared protocol and locked asymmetric settings.

Quantity	Setting
Nominal coverage / decision floor	0.90 / 0.88
History / forecast window	12 / 12 five-minute bins
Training / fault seeds	11, 22, 33 / 101, 202, 303
Scale floor	0.44704 $\mathrm{m\,s^{-1}}$
Buffer / age decay / stale decay	2016 / 288 / 288 bins
Context bandwidth / support reference	2 / 50
Inflation coefficients / cap	0.05, 0.05 / 3
Real local archive cap	256 times group size
Maximum local neighbours	2
Bootstrap resamples	2000

Per-backbone training configurations are supplied in the release. These locked calibration settings supersede exploratory defaults.

7.2 Reporting conditions, controls and stress tests

Table 11 defines the six reporting conditions. C3 combines correlated input outages with delayed feedback and backlog release. C4 imposes permanent feedback loss within the affected region. The baseline connected outage affects 10% of sensors for six bins, with two events per day and a three-bin backlog release. C4 loses half of the within-outage feedback. Physical target validity remains unchanged.

Table 11 Reporting conditions.

Condition	Predictor input	Calibration feedback
C0	Available	Available
C1	Available	Correlated delay
C2	Correlated outage	Available
C3	Correlated outage	Delay and backlog
C4	Correlated outage	Permanent within-outage loss
C5	Independent packet loss	Independent packet loss

C3/C4 baseline: 10% connected sensors, 30-min outage, two daily events, 15-min backlog; C4 permanently loses half of affected feedback. Schedules are separate from physical validity.

The matched-rate independent control randomises affected sensor identities while retaining severity. It asks whether a method succeeds under the independent process; it does not by itself estimate a causal correlated-minus-independent treatment effect. The one-factor stress grid changes duration, connected fraction, additional backlog delay or within-outage loss while holding other settings fixed. Two further tests impair initial calibration and introduce additional low-speed feedback loss. The latter uses a training-derived 20th-percentile speed threshold and suppresses qualifying outcomes with additional probability 0.5. Truth enters only the evaluator-side fault generator. This is missing-not-at-random (MNAR) sensitivity, not an identifiable correction procedure.

7.3 Endpoints and paired inference

For interval $[L, U]$, outcome y and miscoverage α, the interval score [26] is

$$\mathrm{IS}_\alpha(L, U; y) = U - L + \frac{2}{\alpha}(L - y)\mathbf{1}(y < L) + \frac{2}{\alpha}(y - U)\mathbf{1}(y > U). \tag{13}$$

Lower scores reward narrow intervals while penalising missed targets. Coverage is the fraction of originally valid targets inside the interval; width is $U - L$. The primary endpoint averages the 90% score equally over 15-, 30- and 60-minute horizons. Paired differences are comparator minus FAR-Cal, so positive effects favour FAR-Cal. The registered rule requires lower 95% bounds above zero against both rolling and ACI, together with aggregate coverage at least 0.88. Nominal coverage remains 0.90; the 0.88 floor is an evaluation tolerance, not a revised nominal target.

SUMO inference averages model and fault seeds within each episode, then resamples complete episodes 2,000 times. Real-data inference averages model seeds within each fault seed and uses one-day moving blocks, stratified by fault seed. Model-seed repetitions are not independent temporal replications. Real-data audits use systematically selected hourly origins and affected sensors through 12 post-event bins; the buffer cap controls cost. Thus the real-data results do not represent every five-minute

network-wide forecast. Stress summaries average origin-level scores, whereas descriptive runtime quality uses pooled valid targets; these estimands are not substituted for one another.

Seven component ablations retain the same point and scale caches: no spatial pooling, no context matching, no target-age weighting, no stale blend, no inflation, arrival-time freshness and frozen calibration only. The arrival-time diagnostic changes age weighting and freshness gap but retains target-based buffer eligibility and timestamp-block support. Frozen-only uses the same archive and fallback without live adaptation or inflation. Full-reference runs are verified and reused. Ablation intervals are exploratory and unadjusted for 42 comparisons; their signs are descriptive evidence rather than family-wise confirmatory tests.

8 Comparative results

8.1 Development history and fresh validation

The initially locked symmetric formulation failed held-out SUMO C3 testing. A regularised revision had favourable estimates in a 30-episode replication, but both paired intervals included zero. A separately registered 160-episode confirmation again did not establish superiority; comparator advantages were approximately $0.012\,\mathrm{m\,s^{-1}}$ after unit conversion. These unsuccessful stages are retained rather than reclassified as successful evidence for the final method.

The subsequent signed-tail candidate was selected using split-safe tuning and locked before its fresh validation. Table 12 reports the six final FAR-GW dataset/scenario cells. The 40-episode SUMO C3 validation passes the complete rule. SUMO C4 has positive paired score effects but coverage 0.8793, narrowly below the registered floor. Neither favourable score estimates alone nor rounding coverage to two decimals changes that decision.

Table 12 Final FAR-GW paired score effects and joint decisions.

Data	Cell	Rolling minus FAR	ACI minus FAR	Coverage	Pass
SUMO	C3	0.266 [0.105, 0.450]	0.259 [0.104, 0.427]	0.8890	Yes
SUMO	C4	0.460 [0.272, 0.682]	0.422 [0.256, 0.614]	0.8793	No
METR-LA	C3	0.355 [0.192, 0.551]	0.345 [0.185, 0.538]	0.8985	Yes
METR-LA	C4	0.425 [0.238, 0.638]	0.425 [0.253, 0.617]	0.8958	Yes
PEMS-BAY	C3	-0.035 [-0.101, 0.047]	-0.042 [-0.111, 0.043]	0.9006	No
PEMS-BAY	C4	-0.119 [-0.194, -0.043]	-0.130 [-0.216, -0.047]	0.9068	No

Effects and 95% paired intervals are in $\mathrm{m\,s^{-1}}$. Positive differences favour FAR-Cal. SUMO values are rounded from saved validation summaries.

METR-LA passes in both C3 and C4. PEMS-BAY C3 has small effects whose intervals include zero; PEMS-BAY C4 favours both comparators. These negative transfer results are principal findings. The method is left locked rather than retuned using the evaluated test outcomes.

8.2 Cross-backbone and relational comparisons

Fig. 3 presents every evaluated real-data comparator with the proposed method as the zero reference. Within each backbone, all calibrators use the same forecasts and masks. The plot therefore compares calibration conditional on a backbone; it is not a ranking of the predictors themselves. CoRel points appear only for FAR-GW because that is the evaluated scope. Table 13 reports the direct CoRel comparison in numerical form, including both methods' coverage and the paired score interval. All four CoRel-minus-FAR-Cal intervals are positive, but this does not overturn PEMS-BAY failures against rolling and ACI. Different comparators support different conclusions.

Table 13 Direct CoRel adaptation comparison on FAR-GW caches.

Data	Cell	FAR score	CoRel score	CoRel minus FAR	FAR cov.	CoRel cov.
METR-LA	C3	9.091	9.850	0.758 [0.518, 1.030]	0.8985	0.8743
METR-LA	C4	10.476	11.449	0.973 [0.616, 1.403]	0.8958	0.8725
PEMS-BAY	C3	4.617	5.625	1.008 [0.814, 1.215]	0.9006	0.8837
PEMS-BAY	C4	4.723	5.726	1.004 [0.787, 1.227]	0.9068	0.8743

Dimensional scores and paired intervals are in $\mathrm{m\,s^{-1}}$. Absolute scores and CoRel coverage retain the report's rounding; paired effects use full saved precision. This is not a comparison on other backbones.

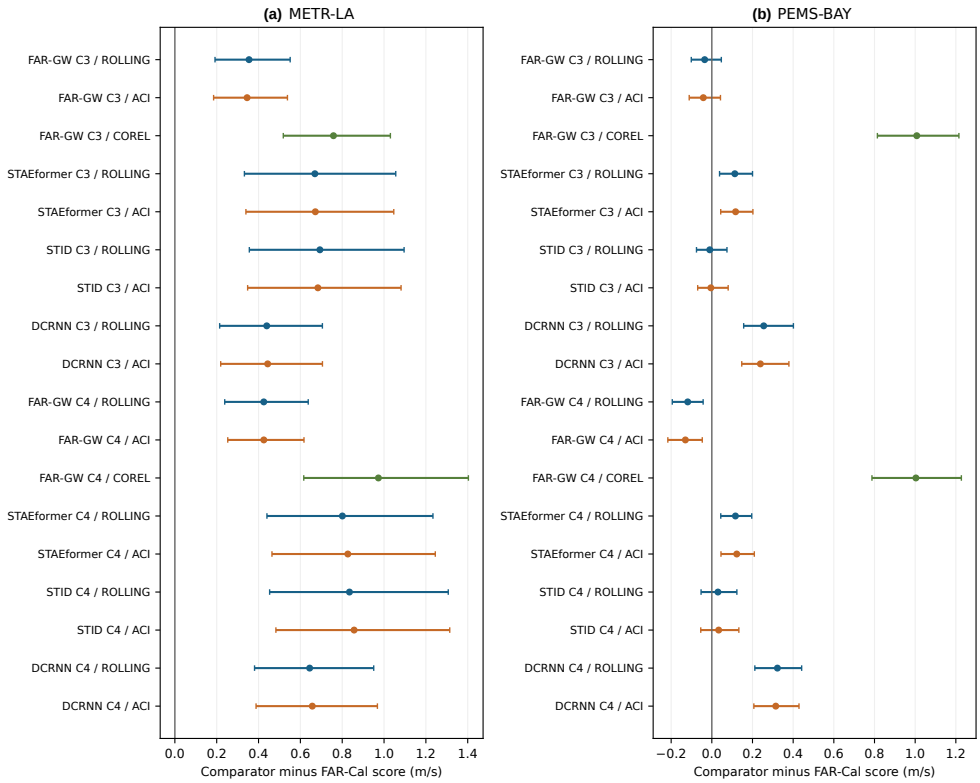


Fig. 3 Paired comparator-minus-FAR-Cal interval-score effects and 95% intervals in SI units. Positive values favour FAR-Cal. Comparisons share frozen forecasts within each backbone; CoRel is restricted to FAR-GW. Panels (a) and (b) show METR-LA and PEMS-BAY.

The four-backbone matrix passes in 12 of 16 registered cells: all eight METR-LA cells, and the four PEMS-BAY cells using STAEformer or DCRNN. FAR-GW and STID each fail both PEMS-BAY conditions. Table 14 provides both paired score effects, coverage and joint decisions for every cell. Its effect columns compare calibrators within a backbone; they must not be read as point-model rankings. Consistency across METR-LA predictors supports transfer on that network, while PEMS-BAY demonstrates predictor-dependent behaviour.

Table 14 All sixteen paired backbone comparisons.

Backbone	Data	Cell	Rolling minus FAR	ACI minus FAR	Coverage	Pass
FAR-GW	METR-LA	C3	0.355 [0.192, 0.551]	0.345 [0.185, 0.538]	0.8985	Yes
STAEformer	METR-LA	C3	0.669 [0.332, 1.056]	0.672 [0.340, 1.047]	0.8965	Yes
STID	METR-LA	C3	0.693 [0.356, 1.096]	0.684 [0.348, 1.081]	0.8891	Yes
DCRNN	METR-LA	C3	0.439 [0.214, 0.705]	0.444 [0.219, 0.706]	0.8891	Yes
FAR-GW	METR-LA	C4	0.425 [0.238, 0.638]	0.425 [0.253, 0.617]	0.8958	Yes
STAEformer	METR-LA	C4	0.801 [0.440, 1.234]	0.827 [0.464, 1.245]	0.8916	Yes
STID	METR-LA	C4	0.835 [0.453, 1.307]	0.857 [0.483, 1.314]	0.8880	Yes
DCRNN	METR-LA	C4	0.644 [0.381, 0.951]	0.657 [0.388, 0.968]	0.8943	Yes
FAR-GW	PEMS-BAY	C3	-0.035 [-0.101, 0.047]	-0.042 [-0.111, 0.043]	0.9006	No
STAEformer	PEMS-BAY	C3	0.113 [0.038, 0.200]	0.117 [0.044, 0.202]	0.8945	Yes
STID	PEMS-BAY	C3	-0.010 [-0.075, 0.074]	-0.005 [-0.070, 0.080]	0.9002	No
DCRNN	PEMS-BAY	C3	0.255 [0.157, 0.401]	0.239 [0.147, 0.379]	0.9007	Yes
FAR-GW	PEMS-BAY	C4	-0.119 [-0.194, -0.043]	-0.130 [-0.216, -0.047]	0.9068	No
STAEformer	PEMS-BAY	C4	0.116 [0.044, 0.196]	0.123 [0.045, 0.209]	0.9101	Yes
STID	PEMS-BAY	C4	0.030 [-0.053, 0.123]	0.034 [-0.054, 0.133]	0.9062	No
DCRNN	PEMS-BAY	C4	0.323 [0.212, 0.442]	0.314 [0.207, 0.429]	0.9144	Yes

Effects and 95% paired intervals are in $\mathrm{m\,s^{-1}}$. Every comparison conditions on one frozen backbone. Joint rule requires both lower bounds above zero and FAR-Cal coverage at least 0.88.

CoRel-minus-FAR-Cal differences are positive in all four FAR-GW real-data cells. After SI conversion, the estimates are 0.758, 0.973, 1.008 and 1.004 $\mathrm{m\,s^{-1}}$ for METR-LA C3/C4 and PEMS-BAY C3/C4. Their respective intervals are [0.518, 1.030], [0.616, 1.403], [0.814, 1.215] and [0.787, 1.227] $\mathrm{m\,s^{-1}}$. CoRel coverage is 0.8743, 0.8725, 0.8837 and 0.8743. This direct comparison supports an advantage over this adaptation; it does not imply that FAR-Cal beats rolling or ACI in the PEMS-BAY cells.

8.3 Coverage and score must be read jointly

Fig. 4 places each final FAR-GW cell's coverage beside its interval score, using identical axes within the quality panel. It makes the SUMO C4 decision visible: positive score improvements coexist with coverage below 0.88. PEMS-BAY coverage exceeds the floor, yet superiority against rolling and ACI is not established. These are different failure modes. Table 12 therefore reports the conjunction, while the figure preserves each metric separately. The vertical separation of network scores is descriptive and not a between-network treatment effect.

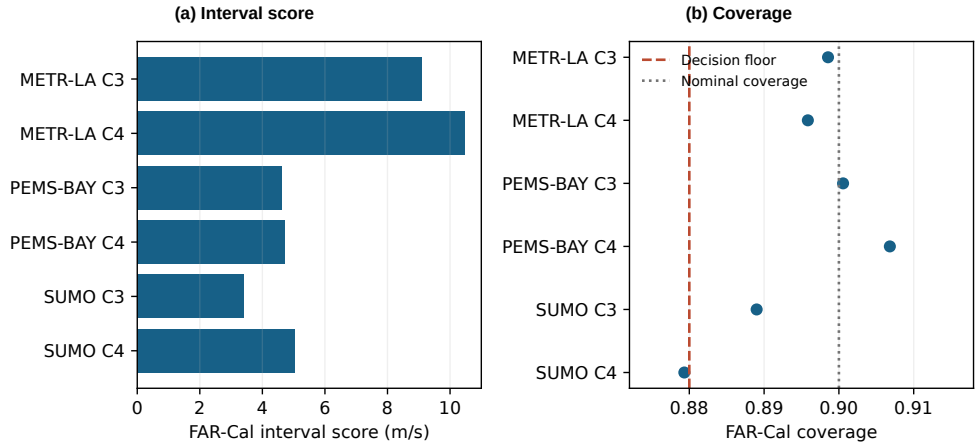


Fig. 4 Final FAR-GW interval score and coverage for the six primary cells. Dashed and dotted lines mark the 0.88 decision floor and 0.90 nominal coverage. Absolute network scores are not a paired between-network effect. Panels (a) and (b) show interval score and coverage.

8.4 Core faults, independent controls and one-factor stress

The core and matched-control matrices complete 168 and 120 stages. Table 15 reports their paired effects and joint decisions. METR-LA passes C1, C2 and C5 and both independent C3/C4 controls; PEMS-BAY passes none. The independent-control results preserve the network-specific pattern without proving that spatial correlation itself causes the method's advantage.

Table 15 Core and independent matched-rate controls.

Data	Cell	Schedule	Rolling effect	ACI effect	Pass
METR-LA	C1	Core	0.356 [0.194, 0.553]	0.347 [0.187, 0.540]	Yes
METR-LA	C2	Core	0.354 [0.192, 0.551]	0.345 [0.185, 0.538]	Yes
METR-LA	C5	Core	0.102 [0.063, 0.129]	0.092 [0.055, 0.120]	Yes
METR-LA	C3	Independent	0.304 [0.225, 0.386]	0.310 [0.232, 0.390]	Yes
METR-LA	C4	Independent	0.307 [0.227, 0.389]	0.312 [0.234, 0.392]	Yes
PEMS-BAY	C1	Core	-0.035 [-0.101, 0.046]	-0.042 [-0.111, 0.042]	No
PEMS-BAY	C2	Core	-0.035 [-0.101, 0.047]	-0.042 [-0.111, 0.043]	No
PEMS-BAY	C5	Core	-0.014 [-0.021, -0.005]	-0.016 [-0.023, -0.007]	No
PEMS-BAY	C3	Independent	0.018 [-0.010, 0.062]	0.018 [-0.010, 0.063]	No
PEMS-BAY	C4	Independent	0.017 [-0.009, 0.055]	0.017 [-0.008, 0.055]	No

Effects and paired 95% intervals are in $\mathrm{m\,s^{-1}}$; the independent controls randomise sensor identities at matched severity.

All 1,488 one-factor stress stages finish, yielding 48 dataset/scenario/variant summaries. METR-LA passes all 24 tested cells and PEMS-BAY none of its 24; every aggregate coverage is at least 0.88. Table 16 groups the findings by factor, and Fig. 5

shows score sensitivity to outage duration. Duration, connected fraction, backlog delay, permanent loss and initial archive impairment test different limits of the reporting system; they are not pooled into a single generalisation guarantee. The MNAR grid retains the same distinction: favourable performance in one network and failure in the other remain compatible with selective outcome observation.

Table 16 One-factor stress summary by network and factor.

Data	Factor	Cells	Passes	Min coverage
METR-LA	duration_bins	6	6	0.8956
METR-LA	connected_fraction	4	4	0.8908
METR-LA	additional_backlog_delay_bins	4	4	0.8959
METR-LA	loss_within	6	6	0.8939
METR-LA	cold_calibration	2	2	0.8959
METR-LA	mnar_low_speed	2	2	0.8959
PEMS-BAY	duration_bins	6	0	0.8981
PEMS-BAY	connected_fraction	4	0	0.9003
PEMS-BAY	additional_backlog_delay_bins	4	0	0.9087
PEMS-BAY	loss_within	6	0	0.8990
PEMS-BAY	cold_calibration	2	0	0.9006
PEMS-BAY	mnar_low_speed	2	0	0.9013

Cells comprise separately evaluated values and C3/C4 conditions; the release retains every effect estimate and interval. MNAR is a sensitivity experiment, not an identification result.

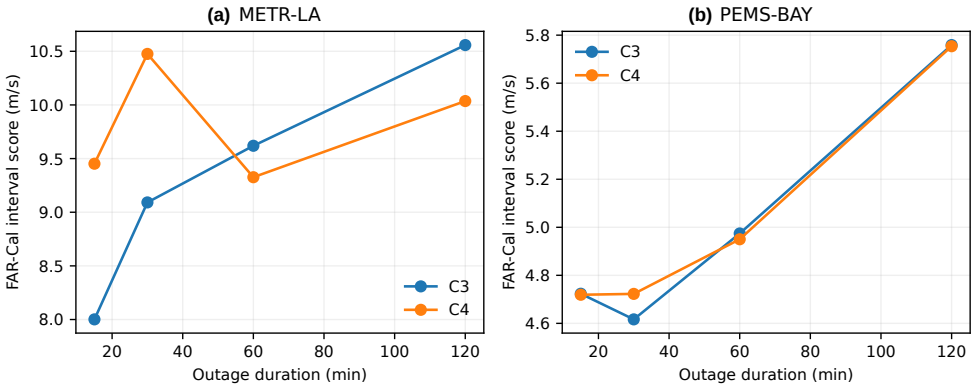


Fig. 5 Interval score across the registered outage-duration grid. Lines connect descriptive scores, not interpolation guarantees; C3 and C4 are distinct feedback conditions. Panels (a) and (b) show METR-LA and PEMS-BAY.

8.5 Additional reporting stress

Fig. 6 displays affected fraction, extra backlog delay, permanent-loss probability, cold calibration and low-speed MNAR sensitivity. Each point compares ACI with FAR-Cal using the registered paired inference, with positive values favouring FAR-Cal. Continuous-factor lines connect tested values only; cold calibration and MNAR are separate categorical experiments. All METR-LA stress cells pass the joint rule, whereas no PEMS-BAY stress cell does. Table 16 groups these decisions; the plotted ACI contrast alone is not the full two-comparator rule. The cold archive tests whether an initially impaired calibration source changes performance, and the MNAR panel tests selective outcome suppression without claiming correction.

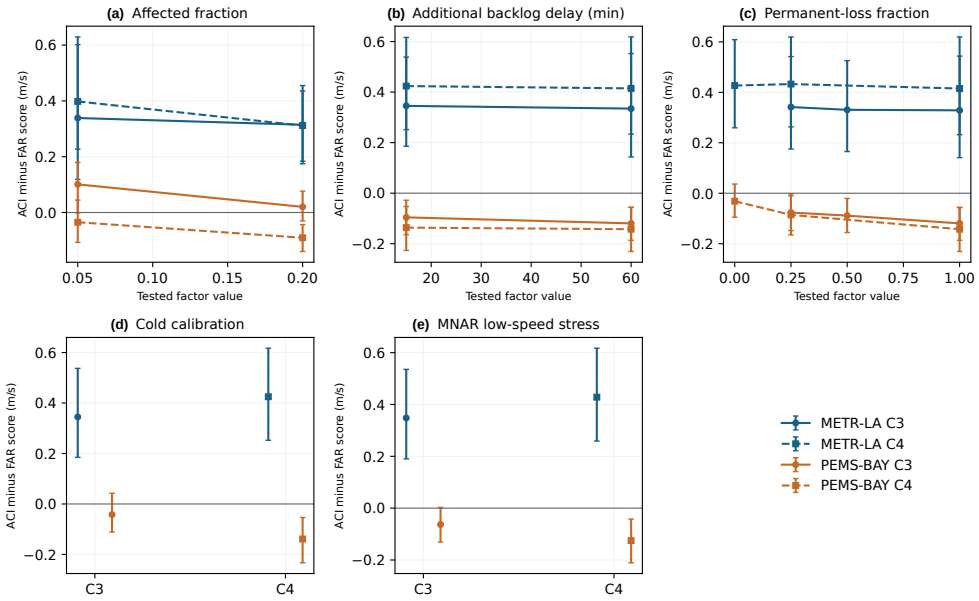


Fig. 6 Additional one-factor stress: ACI-minus-FAR-Cal effects and paired 95% intervals for affected fraction, backlog delay, permanent loss, cold calibration and MNAR. Positive values favour FAR-Cal. Categorical panels contain only their registered conditions; continuous lines connect tested values without extrapolation. Panels (a)–(e) show affected fraction, backlog delay, permanent loss, cold calibration and MNAR, respectively.

The duration curve in Fig. 5 reports absolute FAR-Cal score rather than a paired treatment effect. Longer outages can change which targets enter the fault-focused audit, so the curve cannot isolate a duration effect on a fixed target population. It complements the paired stress results by showing the magnitude of interval-score penalties in each tested condition. These plots are based on completed evaluations, not simulated illustrative values or interpolated predictions of untested fault levels.

9 Full component ablation study

All 432 stages are complete: 54 verified full references and 378 new component evaluations. Fig. 7 displays ablation-minus-full effects. A positive interval indicates degradation after removal; a negative interval indicates an improvement. Thirteen exploratory intervals are wholly positive, four wholly negative and 25 include zero. Table 17 identifies which components fall into these categories.

Table 17 Exploratory paired ablation classifications across six cells.

Removal/diagnostic	Worsens	Improves	Inconclusive
no spatial	3	0	3
no context	1	1	4
no age	0	3	3
no stale blend	5	0	1
no inflation	1	0	5
arrival freshness	2	0	4
frozen only	1	0	5

Worsens: lower 95% bound above zero; improves: upper bound below zero. The intervals are exploratory and unadjusted for multiple comparisons. Counts refer to removals, not full-method superiority.

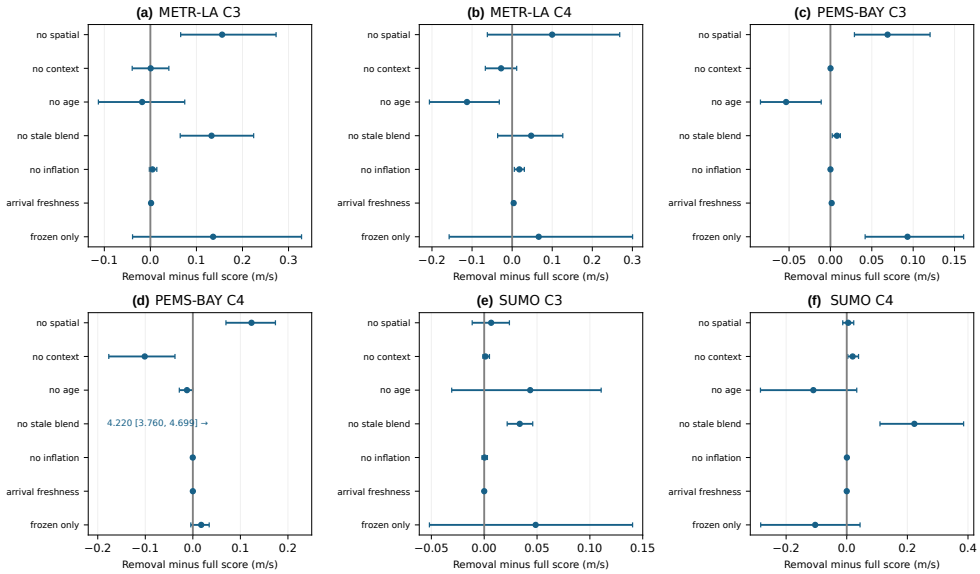


Fig. 7 Exploratory removal-minus-full interval-score effects and 95% paired intervals. Positive values indicate deterioration after removal. Panel scales differ; the off-scale PEMS-BAY C4 stale-blend failure is explicitly labelled with its estimate and bounds. Panels (a)–(f) show METR-LA C3, METR-LA C4, PEMS-BAY C3, PEMS-BAY C4, SUMO C3 and SUMO C4.

Stale blending has the clearest repeated benefit. Removing it worsens score in METR-LA C3, both PEMS-BAY conditions and both SUMO conditions. The largest effect occurs in PEMS-BAY C4: removal increases score by $4.220\,\mathrm{m\,s^{-1}}$, with 95% interval $[3.760, 4.699]\,\mathrm{m\,s^{-1}}$, while coverage falls to 0.549. Spatial pooling is beneficial in METR-LA C3 and PEMS-BAY C3/C4, but its remaining intervals include zero.

The adverse findings matter equally. Removing age weighting improves score in METR-LA C4 and PEMS-BAY C3/C4; removing context matching improves PEMS-BAY C4. Consequently, the ablations do not support a claim that every component is necessary or uniformly beneficial. Arrival-time freshness is worse in METR-LA C4 and PEMS-BAY C3, but the remaining effects are small or uncertain. The simulated diagnostic does not establish a large general backlog penalty. Exploratory intervals are not used to modify the tested method.

Tables 18–20 provide all forty-two numerical removal effects, exploratory intervals, coverage and widths. Table 18 shows that removing age weights improves METR-LA C4 score, while other small effects remain inconclusive. Table 19 shows age-weight improvements in both PEMS-BAY cells and a context-removal improvement in C4. The same table documents the severe stale-blend failure in PEMS-BAY C4. Table 20 shows a useful stale blend in both SUMO conditions, with several remaining intervals spanning zero. These findings are more informative than treating every component as necessary.

Table 18 Full METR-LA component ablations.

Cell	Removal / diagnostic	Removal minus full (95% interval)	Coverage	Width
C3	no spatial	0.156 [0.066, 0.273]	0.8904	4.945
C3	no context	0.001 [-0.039, 0.040]	0.8996	4.919
C3	no age	-0.018 [-0.113, 0.075]	0.9025	4.875
C3	no stale blend	0.133 [0.065, 0.224]	0.8991	5.003
C3	no inflation	0.005 [-0.002, 0.014]	0.8980	4.867
C3	arrival freshness	0.002 [-0.002, 0.004]	0.8985	4.886
C3	frozen only	0.136 [-0.039, 0.328]	0.8835	4.480
C4	no spatial	0.100 [-0.062, 0.268]	0.8889	5.241
C4	no context	-0.028 [-0.067, 0.011]	0.8964	5.186
C4	no age	-0.113 [-0.206, -0.032]	0.9000	5.142
C4	no stale blend	0.047 [-0.036, 0.126]	0.8962	5.313
C4	no inflation	0.018 [0.005, 0.030]	0.8935	5.105
C4	arrival freshness	0.004 [0.002, 0.006]	0.8960	5.166
C4	frozen only	0.066 [-0.157, 0.300]	0.8797	4.683

Effects and widths are in $\mathrm{m\,s^{-1}}$. Positive differences mean removal worsens score. Intervals are exploratory and unadjusted. The frozen-only and arrival-freshness rows are diagnostic alternatives.

Table 19 Full PEMS-BAY component ablations.

Cell	Removal / diagnostic	Removal minus full (95% interval)	Coverage	Width
C3	no spatial	0.069 [0.029, 0.120]	0.8941	2.825
C3	no context	0.000 [-0.001, 0.001]	0.9008	2.799
C3	no age	-0.054 [-0.085, -0.011]	0.9054	2.795
C3	no stale blend	0.008 [0.002, 0.012]	0.8990	2.796
C3	no inflation	0.000 [-0.000, 0.000]	0.9003	2.797
C3	arrival freshness	0.001 [0.001, 0.003]	0.9002	2.796
C3	frozen only	0.093 [0.042, 0.161]	0.9076	2.871
C4	no spatial	0.123 [0.070, 0.174]	0.8999	3.129
C4	no context	-0.101 [-0.177, -0.038]	0.9015	2.978
C4	no age	-0.012 [-0.028, -0.000]	0.9075	3.031
C4	no stale blend	4.221 [3.760, 4.699]	0.5486	1.677
C4	no inflation	-0.000 [-0.003, 0.003]	0.9044	2.999
C4	arrival freshness	0.000 [-0.000, 0.000]	0.9066	3.029
C4	frozen only	0.018 [-0.004, 0.034]	0.9053	3.027

Effects and widths are in $\mathrm{m\,s^{-1}}$. Positive differences mean removal worsens score. Intervals are exploratory and unadjusted. The frozen-only and arrival-freshness rows are diagnostic alternatives.

Table 20 Full SUMO component ablations.

Cell	Removal / diagnostic	Removal minus full (95% interval)	Coverage	Width
C3	no spatial	0.007 [-0.011, 0.024]	0.8875	2.082
C3	no context	0.001 [-0.001, 0.005]	0.8889	2.080
C3	no age	0.044 [-0.031, 0.111]	0.8994	2.290
C3	no stale blend	0.034 [0.022, 0.046]	0.8817	2.043
C3	no inflation	0.000 [-0.002, 0.003]	0.8878	2.071
C3	arrival freshness	0.000 [-0.000, 0.000]	0.8890	2.081
C3	frozen only	0.049 [-0.052, 0.140]	0.9041	2.362
C4	no spatial	0.005 [-0.013, 0.023]	0.8780	2.407
C4	no context	0.019 [0.003, 0.039]	0.8790	2.422
C4	no age	-0.110 [-0.285, 0.033]	0.8915	2.559
C4	no stale blend	0.223 [0.110, 0.386]	0.8579	2.332
C4	no inflation	0.000 [-0.003, 0.004]	0.8781	2.405
C4	arrival freshness	0.000 [-0.000, 0.000]	0.8793	2.416
C4	frozen only	-0.104 [-0.284, 0.044]	0.8961	2.617

Effects and widths are in $\mathrm{m\,s^{-1}}$. Positive differences mean removal worsens score. Intervals are exploratory and unadjusted. The frozen-only and arrival-freshness rows are diagnostic alternatives.

Fig. 7 uses separate panel scales to preserve small effects while explicitly labelling the off-scale PEMS-BAY C4 stale-blend difference. Its zero line corresponds to no paired score change. A beneficial removal is evidence against claiming that the removed component uniformly improves quality. Conversely, an inconclusive interval does not prove redundancy. Dependence between components, limited episode/day replication and multiple exploratory comparisons prevent a simple causal ranking of design choices. The locked full method is not retrospectively retuned to maximise these test-set results.

9.1 Descriptive archive-cap sensitivity

Eighteen timed replays compare archive caps 64, 128 and 256 with identical C3 inputs and fixed model/fault seeds 11/101. Table 21 reports wall-clock medians, observed ranges and pooled quality. Timing includes central processing unit (CPU) calibration, loading and output serialisation, excluding training and prediction. The cap-256 setting exactly reproduces the archived same-run quality. Smaller caps have slightly higher score in both fixed runs, but timing ranges overlap. Host workload and filesystem cache are not isolated, so these measurements do not establish a reliable speed advantage, a cross-method runtime ranking or a deployment throughput guarantee.

Table 21 Descriptive CPU archive-cap replays.

Data	Cap	Median seconds [range]	Score ($m\,s^{-1}$)	Coverage
METR-LA	64	19.28 [19.25, 19.89]	9.4872	0.9114
METR-LA	128	19.69 [18.71, 20.04]	9.3233	0.9133
METR-LA	256	19.08 [18.80, 19.99]	9.3148	0.9122
PEMS-BAY	64	97.95 [96.88, 99.69]	5.4284	0.8980
PEMS-BAY	128	93.76 [91.85, 102.98]	5.4079	0.8971
PEMS-BAY	256	96.25 [91.57, 105.95]	5.3960	0.8971

Three repetitions per cap and network; identical C3 duration03 inputs, seeds 11/101. Ranges are observed timing ranges, not confidence intervals. The pooled quality estimand differs from origin-level paired inference.

Fig. 8 complements Table 21: horizontal ranges show three observed replay times at each cap, and colour/labels identify the cap rather than another forecasting method. The score axis uses the same-run pooled quality estimand. No error bar is a runtime confidence interval. This figure is a reproducibility and resource sensitivity diagnostic, not a speed competition among calibrators or evidence of service-level latency.

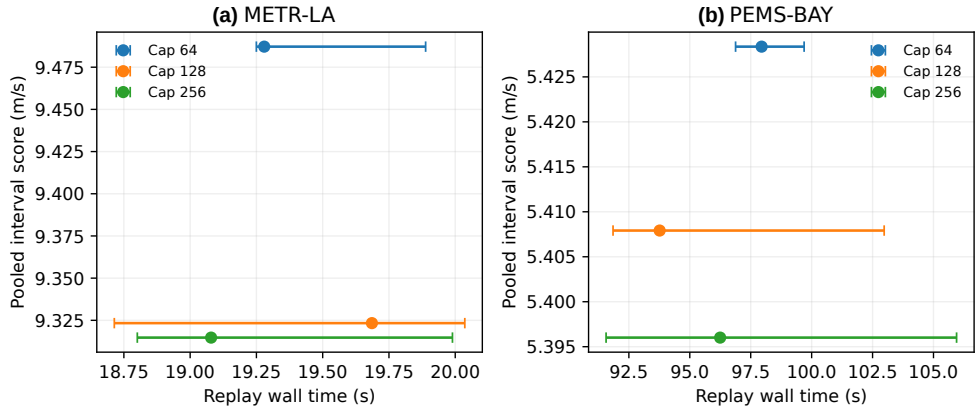


Fig. 8 Descriptive archive-cap quality and wall-time sensitivity, with three observed runtime repetitions per cap and network. Horizontal bars are observed ranges, not confidence intervals. These fixed-run pooled scores differ from the origin-level paired inference estimand. Panels (a) and (b) show METR-LA and PEMS-BAY.

10 Discussion

10.1 Interpretation and methodological limits

The strongest methodological lesson is that observing an outcome and receiving its report are different events. Calibration needs the latter, while valid evaluation uses the former. Keeping these channels separate exposes a failure mode that input-only

missingness experiments cannot fully characterise: the forecasting system can lose both explanatory observations and the evidence needed to correct its uncertainty.

The results favour a conditional interpretation. The asymmetric redesign succeeds on fresh SUMO C3 and transfers consistently across METR-LA backbones, while PEMS-BAY remains heterogeneous. Residual shape, graph neighbourhoods, scale quality and network dynamics may contribute, but the experiments do not isolate a dominant explanation. The failed symmetric confirmations also caution against attributing all final gains to a single component change: successive formulations were evaluated on different held-out stages.

Full ablations sharpen this conclusion. Stale blending is repeatedly useful, and its removal under permanent loss can collapse coverage. Conversely, target-age and context weighting sometimes hurt score. These findings suggest a future programme of training/tuning-only component selection followed by validation on new networks or untouched episode sets. Choosing a revised rule from the current test ablations would be further method development, not confirmation of the locked rule.

Neither weighted pooling nor the support statistic supplies a missing-outcome identification argument. Selective feedback can change the residual population, especially if low speeds are preferentially suppressed. The MNAR experiment quantifies sensitivity to a specified fault generator; it cannot recover the distribution of unseen outcomes without assumptions or additional information. There is no claim of identifiable MNAR correction or finite-sample coverage under arbitrary reporting failures.

10.2 Motivation, societal implications and deployment boundary

The practical aim is to make uncertainty services more accountable when sensing infrastructure degrades. Wider or better-positioned intervals may help operators recognise that a point forecast is less informative during an outage. That possibility is a motivation, not a demonstrated improvement in routing, travel time, safety, emissions or public welfare. No traffic-control policy is evaluated in this study.

Uneven detector availability also raises distributional concerns. Regions with persistently poor instrumentation may receive intervals built from neighbouring or historical residuals that do not represent local conditions. Aggregate network coverage can hide such disparities. A deployment should therefore audit coverage and interval quality by location, reporting quality and relevant operational regime, communicate outages explicitly and retain human oversight for consequential actions. The present datasets and simulation are insufficient to claim equitable performance across communities or road types.

The architecture records forecast issuance and subsequent feedback, which can support auditing. A production implementation would additionally require access controls, retention policies, operational monitoring and validation on live infrastructure. The existing gateway tests establish replay-equivalence properties at smoke scale; they do not replace a full service study. This manuscript accordingly claims a causal replay benchmark and tested calibration behaviour, not operational readiness.

10.3 Limitations

The real-data audit is hourly, fault focused and locally capped. Its conclusions cannot be substituted for complete five-minute network-wide evaluation. Simulation provides controlled replication but represents a small freeway network. CoRel, STID and DCRNN are documented adaptations, and STAEformer is an independent implementation; the comparisons do not reproduce every source-paper setting. The archive-cap study uses one fixed run per network and descriptive timing. Ablation intervals are exploratory and not adjusted for multiple comparisons. Dataset redistribution rights for the exact real benchmark packages remain unverified, so the release provides acquisition instructions rather than redistributing them. Finally, the literature positioning is bounded: no universal absence of related feedback-aware work is asserted.

10.4 What the comparisons establish

The backbone comparison and the component study answer complementary questions. Table 14 shows that a method can transfer across forecasting representations without performing uniformly across networks. Tables 18–20 show that a useful full pipeline can contain components that are adverse in specific cells. Together they argue for conditional rather than universal recommendations. The experiments do not identify whether scale quality, residual asymmetry, neighbourhood heterogeneity or temporal dynamics drives the PEMS-BAY interaction. Establishing that mechanism would require an additional registered comparison, not a post hoc explanation presented as fact.

The fresh C3 simulation validation offers protection against selecting a candidate from its own validation outcomes. C4 transfer is a separate assessment on the existing held-out set and misses the coverage criterion. These distinctions matter because increasing the number of model or fault seeds does not create new independent physical episodes. Likewise, the 432 ablation stages are an execution count, not 432 independent inferential observations. The paired resampling units remain episodes in simulation and day blocks within fault seeds in the real audit.

10.5 Relationship to recent feedback theory

Published online and multivariate conformal studies reinforce the importance of specifying adaptation and dependence assumptions [9, 10, 23]. They also restrict the claim of originality. Our contribution is the traffic-specific executable intervention design, its archive-based signed-residual kernel and its completed controls, transfer and removal evidence. It does not establish a stronger coverage theorem. The published incomplete-time-series study overlaps with the motivation [21]. Only explicit differences in the reviewed protocols are asserted; recognising related objectives does not imply that their methods were evaluated or that their guarantees apply to missing outcome feedback.

10.6 Implications for subsequent studies

A next confirmatory study could preregister network-specific eligibility for age and context weighting, evaluate it on newly held-out networks and compare calibration under equal resource budgets. Sensor-level and fault-regime coverage would test whether aggregate performance hides local failures. A separately registered live study would be required for latency and operational reliability claims, while a traffic-control experiment would be required for travel-time or safety benefits. These proposals follow from observed limitations; none is counted as completed evidence in this manuscript.

11 Conclusion

Separating predictor-input and calibration-feedback availability enables causal assessment of traffic prediction intervals under delayed and permanently lost reports. The locked asymmetric method improves interval quality under several tested conditions, but superiority depends on network and backbone, and the full ablations reveal both useful and adverse components. The study contributes an auditable benchmark and conditional empirical evidence. It does not establish universal superiority, arbitrary-missingness coverage or identifiable MNAR correction.

Acknowledgements. The authors report no additional contributors to acknowledge.

Declarations

Funding. This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Competing interests. The authors declare that they have no competing interests or financial conflicts to disclose.

Ethics approval and consent to participate. Not applicable to the benchmark-data and simulation experiments reported here.

Consent for publication. Not applicable; the manuscript reports no identifiable individual participant data.

Author contributions. Shishir Singh Chauhan conducted the research, implemented the method and experiments, and wrote the manuscript. Satyabrata Roy reviewed the paper and contributed to its writing. Satyabrata Roy is the corresponding author.

Use of generative AI and AI-assisted technologies. During the preparation of this work, the authors used OpenAI Codex to assist with manuscript drafting and editing. The authors reviewed and edited the content and take full responsibility for the published article. The tool also assisted with software implementation, experiment orchestration, artifact checking and figure generation; all numerical findings reported here are drawn from saved evaluation outputs.

Reproducibility, data and code availability. Code, locked configurations, generated SUMO prepared datasets, evidence summaries and manuscript sources are publicly available at https://github.com/shishirbit/feedback-aware-traffic-calibration. The release records seeds, environments, source provenance and artifact hashes. METR-LA and PEMS-BAY must be acquired separately using the links and SHA-256 checks in `docs/sources.md`, because redistribution rights for the exact benchmark packages remain unverified. Training checkpoints and bulky prediction/residual caches must be regenerated before running the evaluation matrices. SUMO episode XML can be regenerated from registered plans; prepared arrays for the original, replication, confirmation and fresh validation sets are included. Negative and inconclusive findings are retained. These resources support reproduction of the specified replay experiments and do not establish production-service readiness.

References

[1] Vlahogianni, E.I., Karlaftis, M.G., Golias, J.C.: Short-term traffic forecasting: Where we are and where we're going. Transportation Research Part C: Emerging Technologies **43**, 3–19 (2014) https://doi.org/10.1016/j.trc.2014.01.005

[2] Khosravi, A., Mazloumi, E., Nahavandi, S., Creighton, D., Lint, J.W.C.: Prediction intervals to account for uncertainties in travel time prediction. IEEE Transactions on Intelligent Transportation Systems **12**(2), 537–547 (2011) https://doi.org/10.1109/TITS.2011.2106209

[3] Wu, Z., Pan, S., Long, G., Jiang, J., Zhang, C.: Graph WaveNet for deep spatial-temporal graph modeling. In: Proceedings of the Twenty-Eighth International Joint Conference on Artificial Intelligence, pp. 1907–1913 (2019). https://doi.org/10.24963/ijcai.2019/264 . https://doi.org/10.24963/ijcai.2019/264

[4] Li, Y., Yu, R., Shahabi, C., Liu, Y.: Diffusion convolutional recurrent neural network: Data-driven traffic forecasting. In: International Conference on Learning Representations (2018). https://openreview.net/forum?id=SJiHXGWAZ

[5] Shao, Z., Zhang, Z., Wang, F., Wei, W., Xu, Y.: Spatial-temporal identity: A simple yet effective baseline for multivariate time series forecasting. In: Proceedings of the 31st ACM International Conference on Information and Knowledge Management, pp. 4454–4458 (2022). https://doi.org/10.1145/3511808.3557702 . https://doi.org/10.1145/3511808.3557702

[6] Liu, H., Dong, Z., Jiang, R., Deng, J., Deng, J., Chen, Q., Song, X.: Spatio-temporal adaptive embedding makes vanilla transformer SOTA for traffic forecasting. In: Proceedings of the 32nd ACM International Conference on Information and Knowledge Management, pp. 4125–4129 (2023). https://doi.org/10.1145/3583780.3615160 . https://doi.org/10.1145/3583780.3615160

[7] Gibbs, I., Candès, E.J.: Adaptive conformal inference under distribution shift.

In: Advances in Neural Information Processing Systems, vol. 34 (2021). https://proceedings.neurips.cc/paper/2021/hash/0d441de75945e5acbc865406fc9a2559-Abstract.html

[8] Hallberg Szabadváry, J.: Adaptive conformal inference for multi-step ahead time-series forecasting online. In: Proceedings of the Thirteenth Symposium on Conformal and Probabilistic Prediction with Applications, vol. 230, pp. 250–263. PMLR (2024). https://proceedings.mlr.press/v230/hallberg-szabadvary24a.html

[9] Gibbs, I., Candès, E.J.: Conformal inference for online prediction with arbitrary distribution shifts. Journal of Machine Learning Research **25**(162), 1–36 (2024)

[10] Schlembach, F., Smirnov, E., Koprinska, I., Winands, M.H.M.: Conformal multistep-ahead multivariate time-series forecasting. Machine Learning **114**(7) (2025) https://doi.org/10.1007/s10994-024-06722-9 . Article 165

[11] Williams, B.M., Hoel, L.A.: Modeling and forecasting vehicular traffic flow as a seasonal ARIMA process: Theoretical basis and empirical results. Journal of Transportation Engineering **129**(6), 664–672 (2003) https://doi.org/10.1061/(ASCE)0733-947X(2003)129:6(664)

[12] Lv, Y., Duan, Y., Kang, W., Li, Z., Wang, F.-Y.: Traffic flow prediction with big data: A deep learning approach. IEEE Transactions on Intelligent Transportation Systems **16**(2), 865–873 (2015) https://doi.org/10.1109/TITS.2014.2345663

[13] Polson, N.G., Sokolov, V.O.: Deep learning for short-term traffic flow prediction. Transportation Research Part C: Emerging Technologies **79**, 1–17 (2017) https://doi.org/10.1016/j.trc.2017.02.024

[14] Cini, A., Marisca, I., Alippi, C.: Filling the g_ap_s: Multivariate time series imputation by graph neural networks. In: International Conference on Learning Representations (2022). https://openreview.net/forum?id=kOu3-S3wJ7

[15] Angelopoulos, A.N., Bates, S.: Conformal prediction: A gentle introduction. Foundations and Trends in Machine Learning **16**(4), 494–591 (2023) https://doi.org/10.1561/2200000101

[16] Lei, J., G'Sell, M., Rinaldo, A., Tibshirani, R.J., Wasserman, L.: Distribution-free predictive inference for regression. Journal of the American Statistical Association **113**(523), 1094–1111 (2018) https://doi.org/10.1080/01621459.2017.1307116

[17] Romano, Y., Patterson, E., Candès, E.J.: Conformalized quantile regression. In: Advances in Neural Information Processing Systems, vol. 32 (2019). https://proceedings.neurips.cc/paper/2019/hash/5103c3584b063c431bd1268e9b5e76fb-Abstract.html

[18] Barber, R.F., Candès, E.J., Ramdas, A., Tibshirani, R.J.: Conformal prediction beyond exchangeability. The Annals of Statistics **51**(2), 816–845 (2023) https://doi.org/10.1214/23-AOS2276

[19] Xu, C., Xie, Y.: Sequential predictive conformal inference for time series. In: Proceedings of the 40th International Conference on Machine Learning, vol. 202, pp. 38707–38727. PMLR (2023). https://proceedings.mlr.press/v202/xu23r.html

[20] Cini, A., Jenkins, A., Mandic, D., Alippi, C., Bianchi, F.M.: Relational conformal prediction for correlated time series. In: Proceedings of the 42nd International Conference on Machine Learning, vol. 267, pp. 10949–10965. PMLR (2025). https://proceedings.mlr.press/v267/cini25a.html

[21] Chen, B., Zhou, X., Cheng, L.: Post-training adaptive conformal prediction for incomplete time series. Transactions on Machine Learning Research (2026)

[22] Jensen, V., Bianchi, F.M., Anfinsen, S.N.: Ensemble conformalized quantile regression for probabilistic time series forecasting. IEEE Transactions on Neural Networks and Learning Systems **35**(7), 9014–9025 (2024) https://doi.org/10.1109/TNNLS.2022.3217694

[23] Zhou, X., Chen, B., Gui, Y., Cheng, L.: Conformal prediction: A data perspective. ACM Computing Surveys **58**(2), 1–37 (2025) https://doi.org/10.1145/3736575 . Article 49

[24] Alvarez Lopez, P., Behrisch, M., Bieker-Walz, L., Erdmann, J., Flötteröd, Y.-P., Hilbrich, R., Lücken, L., Rummel, J., Wagner, P., Wießner, E.: Microscopic traffic simulation using sumo. In: 2018 21st International Conference on Intelligent Transportation Systems, pp. 2575–2582 (2018). https://doi.org/10.1109/ITSC.2018.8569938 . https://doi.org/10.1109/ITSC.2018.8569938

[25] Koenker, R., Bassett, G. Jr.: Regression quantiles. Econometrica **46**(1), 33–50 (1978) https://doi.org/10.2307/1913643

[26] Gneiting, T., Raftery, A.E.: Strictly proper scoring rules, prediction, and estimation. Journal of the American Statistical Association **102**(477), 359–378 (2007) https://doi.org/10.1198/016214506000001437

Author biographies

Shishir Singh Chauhan is affiliated with the Department of Computer Science and Engineering, Manipal University Jaipur, India. His contributions to this study include the research, software implementation, experimental evaluation and manuscript writing.

Satyabrata Roy is affiliated with the Department of Computer Science and Engineering, Manipal University Jaipur, India. He contributed to reviewing and writing the manuscript and serves as its corresponding author.